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Scope Statement: BasilCare AI

Intelligent Cloud of Things Platform for Hydroponic Basil Cultivation

Cloud of Things Project 2026–2027

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1. Concept and Rationale

1.1 Project context

Tunisia faces significant water-resource pressure, making precise irrigation and nutrient management important targets for agricultural innovation. Soilless hydroponic cultivation recirculates nutrient solution and offers a controlled setting in which water inputs, solution quality and crop responses can be monitored. This project proposes an instrumented, small-scale hydroponic farm for sweet basil (*Ocimum basilicum* L.), supported by an IoT gateway, cloud services and intelligent decision support.

1.2 Why basil?

Basil (*Ocimum basilicum*) is a widely used culinary and aromatic herb valued for its fresh leaves. It is a practical crop for a compact hydroponic prototype: its relatively short growth cycle, manageable size and harvestable foliage allow repeated monitoring during a university project. Basil growth depends on stable nutrient-solution conditions, including pH, electrical conductivity and temperature, making it well suited to sensor-based monitoring and alerts. Its leaves also provide visible signs of stress or disease that can motivate camera-based screening, subject to dataset availability and model validation. Published comparisons of hydroponic and aquaponic basil cultivation further support the feasibility of soilless production [3]. Together, these characteristics make basil a relevant test crop for integrating IoT, Edge AI and cloud-based decision support.

1.3 Problem statement

Small hydroponic installations can suffer from unobserved pH / electrical-conductivity (EC) drift, low reservoir level, circulation-pump failures, unsuitable microclimate conditions and delayed recognition of visible leaf abnormalities. Raw readings alone do not necessarily give operators actionable guidance. Meanwhile, cloud-only diagnostics are vulnerable to interrupted connectivity. An integrated monitoring and decision-support platform is proposed to address these gaps.

1.4 Objectives and success criteria

- Acquire, timestamp, store and display environmental and nutrient-solution measurements in near real time.
- Detect configurable anomalies and notify the responsible operator; allow safe pump control under predefined rules.
- Perform lightweight edge-based image screening for visible basil-leaf abnormalities, with uncertainty clearly displayed.
- Provide an evidence-grounded agronomic assistant using RAG and authorized MCP tools.
- Deliver separate farmer and administration PWAs, a documented Jakarta EE 11 REST API and reproducible deployment.

> **Measurement principles:** Water-use reduction and diagnostic quality are intended evaluation outcomes, not guaranteed results. Reference operating ranges will be checked against the selected basil cultivar, nutrient solution and agronomic sources before activation.

2. Clients, Needs, and Functional Scope

2.1 Stakeholders and target users

Stakeholder	Primary need
Small hydroponic growers	Know whether solution quality, water circulation and plant appearance require attention.
Agronomists / educators	Review measurements, explain anomalies and compare interventions across growing cycles.
Farm operators	Receive timely alerts, inspect devices and safely execute approved maintenance actions.
System administrators	Manage users, roles, devices, AI configurations, telemetry quality and incidents.

2.2 Functional requirements

ID	Requirement	Priority
FR-01	Collect ambient temperature/humidity, solution pH/EC, solution temperature and water level.	Must
FR-02	Publish readings over MQTT and visualize live and historical trends.	Must
FR-03	Generate incidents for invalid thresholds, stale readings and device/pump failures.	Must
FR-04	Notify farmer, acknowledge alerts and record their resolution.	Must
FR-05	Run edge image screening and attach confidence/quality indicators to results.	Must
FR-06	Answer documented cultivation questions via RAG with source references.	Must
FR-07	Read sensor history and request approved actions through MCP tools.	Must
FR-08	Operate distinct role-based farmer and administrator PWAs.	Must
FR-09	Report consumption, water level, device uptime and alert frequency.	Must
FR-10	Recommend or perform nutrient dosing autonomously.	Outside MVP

2.3 Exclusions and boundaries

The prototype will not certify plant disease, replace agronomists, guarantee yield improvements, or autonomously dispense chemical nutrients. A laboratory-scale reservoir and limited number of pots are planned; industrial greenhouse control, nationwide rollout, and commercial-grade irrigation certification are beyond the initial scope. Historical data must be clearly separated from live data when connectivity is interrupted.

3. Equipment and Technical Choices

3.1 Prototype equipment (proposed bill of materials)

Component	Role / implementation note
Raspberry Pi 4 Model B (8 GB)	MQTT edge gateway, local buffering, sensor orchestration, lightweight image inference.
Ambient temperature / humidity sensor	Greenhouse microclimate observation (e.g., BME280).
Solution pH probe + interface	Nutrient-solution acidity; requires calibration.
EC probe + interface	Conductivity proxy for dissolved ions; not a sodium-specific detector.
Water-temperature and level sensors	Root-zone monitoring and low-reservoir detection.
Camera module / USB camera	Consistent images of leaves under controlled lighting.
Reservoir, grow tray, tubing and pump	Recirculating hydroponic prototype.
Relay / driver, safe power supply	Low-voltage pump control with electrical isolation and fail-safe logic.

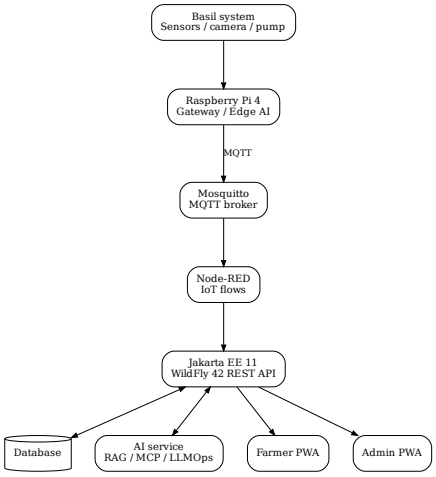
Procurement note: Exact sensor models, costs, wiring, calibration methods and availability will be confirmed during feasibility testing. Camera inference is not assumed to meet real-time frame rates on the Pi.

3.2 Selected software stack

Layer	Technology / responsibility
Edge gateway	Raspberry Pi OS, Python, image-inference runtime, local queue.
Messaging and flows	Mosquitto MQTT broker; Node-RED for ingestion, routing and automation.
Core application	Jakarta EE 11 REST services on WildFly 42.0.0.Final, Java 21.
Persistence	SQL database for users/controls; time-stamped telemetry store (final choice during design).
Web applications	Two installable PWAs: farmer front-office and administrator back-office.
AI services	Separate RAG/LLM service, MCP tool server and evaluation/logging pipeline.
Infrastructure	Cloud IaaS host(s), TLS reverse proxy, GitHub-sourced deployments.

WildFly 42.0.0.Final was released on 22 October 2026 and supports the Jakarta EE 11 platform on Java 21 [4].

4. Logical System Architecture



4.1 Data flow

A sensor adapter collects calibrated measurements on the Pi. MQTT transports telemetry to the broker, Node-RED normalizes and routes events, and Jakarta EE persists validated records and exposes them through role-protected endpoints. The PWAs display readings, alerts and trends. Images are analyzed locally where feasible, and derived observations are sent to the cloud. The AI assistant queries approved MCP operations and the curated RAG corpus, but cannot bypass actuator safety limits.

4.2 Reliability and security

MQTT authentication and access control, TLS at external boundaries, API authorization, audit logs and retained local readings are planned. Critical pump cutoffs remain enforceable locally when the cloud is unavailable. All timestamps use explicit timezone-aware formatting; sensor calibration and offline gaps must be visible to users.

5. AI Integration: Edge AI, RAG, MCP, LLMOps

5.1 Edge AI

A compact image classifier or anomaly screener will run on the Raspberry Pi 4, using controlled leaf images and an appropriate quantized runtime if supported. Initial classes will depend on available labeled basil data. Predictions will include a confidence/uncertainty indicator, image-quality checks and a "review required" fallback. The model will not be described as diagnosing pathogens without suitable ground-truth validation.

5.2 Retrieval-augmented generation

A curated corpus (crop guides, device manuals, calibration procedures and validated operational instructions) will be chunked, embedded and indexed. The assistant will retrieve relevant passages and cite them when explaining pH/EC incidents, maintenance steps or basil stress conditions. Unsupported claims will trigger an explicit uncertainty response.

5.3 Model Context Protocol (MCP)

An MCP server will expose constrained, auditable tools including readings, alert lookup, device status and a pump-action request endpoint. All tools will apply role checks; mutating operations require user confirmation and a deterministic safety-policy check. MCP connects the assistant to tools; it does not replace the Jakarta EE API or MQTT transport.

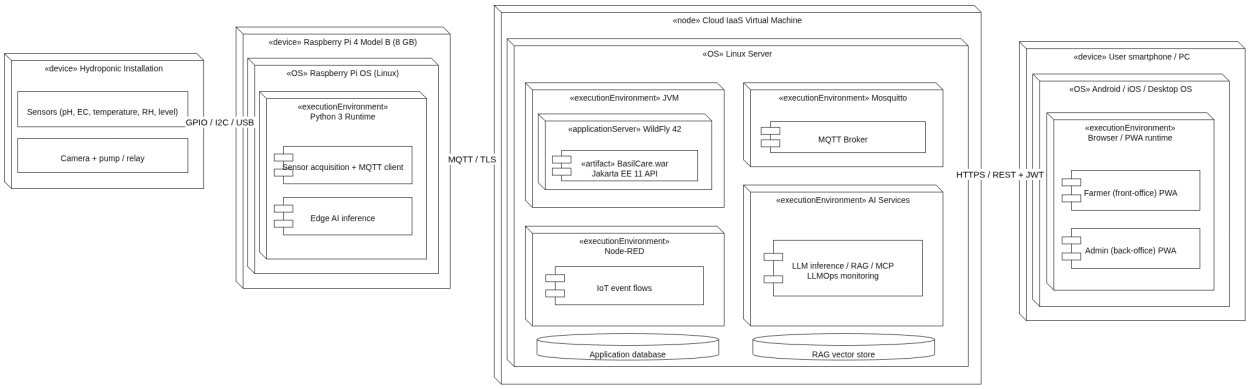
5.4 LLMOps and evaluation

Prompt and model versions, retrieval context IDs, tool-call traces, latency, error rates and evaluation results will be stored with sensitive values minimized. A curated test set will assess groundedness, citation correctness, tool-use authorization and unsafe action refusal. Edge screening will be evaluated separately using confusion matrices and held-out images.



Feasibility decision: The Raspberry Pi hosts sensors and lightweight Edge AI. A separate cloud service hosts the LLM and retrieval stack unless benchmarks demonstrate a viable local alternative.

6. Deployment Diagram



6.1 Deployment constraints

Cloud code will be installed and updated from versioned GitHub repositories (initial `git clone`; subsequent `git pull`), consistent with the professor's instructions. Environment secrets must not be committed. The deployed REST API will have a Postman collection; the two PWAs will be installable on team members' mobile phones. Device authentication and external exposure of MQTT/MCP services will be reviewed before Internet deployment.

6.2 Key non-functional targets (provisional)

- Device measurements should appear on the dashboard within an agreed interval under normal connectivity (target to benchmark).
- Cloud disconnection must not cause unbounded pump operation; offline measurements are queued within local storage limits.
- All user-visible control actions are attributable to a role and timestamp.
- The prototype should be reproducible using documented environment configuration, setup scripts and tests.

7. Project Planning and Methodology

7.1 Seven-week indicative roadmap

Task	W1	W2	W3	W4	W5	W6	W7
Scope and planning	X						
Hardware and calibration	X	X	X				
MQTT and Node-RED		X	X	X			
WildFly API and storage		X	X	X	X		
Two PWAs			X	X	X	X	
Edge AI, RAG and MCP				X	X	X	
Integration and tests					X	X	X
Documentation and demo						X	X

Planning assumptions. The schedule above is indicative, not a professor-announced implementation deadline. It assumes access to the Raspberry Pi and sensors, available development time, and no long procurement delays. The team will assign owners and calendar dates after confirming team size and laboratory availability.

7.2 Preliminary work estimate

Work package	**Estimated person-days**
Requirements, design and UML	5
Hardware, sensor calibration and MQTT	8
Jakarta EE, persistence and authorization	9
Farmer and administrator PWAs	10
Edge AI, RAG, MCP and LLMOps baseline	12
Integration, evaluation, documentation and videos	10
Total indicative effort	**54**

7.3 eXtreme Programming (XP)

Work will be organized into short iterations with user stories, pair programming for complex changes, tests, continuous integration, simple design and frequent stakeholder feedback. Each GitHub push should have a meaningful description and leave compilable code. Initial design artifacts use IoT Thing/Sensor/Actuator abstractions; Observer, Strategy and Command patterns will be evaluated where applicable in the later design notebook.

8. Constraints, Assumptions and Risks

8.1 Project constraints

- **Technology:** Jakarta EE 11, WildFly 42, Node-RED, MQTT, Cloud IaaS, two PWAs, RAG, MCP, LLMOps and Edge AI are required; the Raspberry Pi 4 B (8 GB) is the selected edge device.
- **Resources:** Student budget, sensor calibration materials, limited compute, image-data quality and team availability constrain scope.
- **Methods:** XP, UML artifacts, justified patterns, source documentation, GitHub traceability and individual deliverable videos (maximum five minutes).

8.2 Working assumptions

The prototype will use a small recirculating reservoir and a limited set of basil plants in a controlled indoor or greenhouse-like environment. Team members will have Internet access for deployment and mobile devices for PWA installation. Suitable openly licensed agronomic documents will be available for RAG. Procurement and final infrastructure allocation are not yet confirmed.

8.3 Risk register

Risk	Severity	Mitigation
Poor pH/EC calibration	High	Calibration protocol, reference solutions, validity checks.
Insufficient basil image data	High	Early dataset audit; narrow claim to anomaly screening.
Edge inference too slow	Medium	Benchmark quantized alternatives; lower capture frequency.
Cloud/MQTT outage	High	Local queue, safe default states and offline indication.
LLM unsafe advice or tool usage	High	Verified corpus, authorization, action confirmation, deterministic constraints.
Leaked secrets / unauthorized control	High	Role-based access, TLS, secret storage, audit.
Hardware procurement delays	Medium	Emulated sensors and simulator for parallel backend progress.

8.4 Acceptance and evaluation plan

The demonstration will include: one end-to-end telemetry path; successful alert generation and acknowledgment; an authorized control request with a safety rejection case; installable farmer/admin PWAs; an Edge AI evaluation on a held-out set; RAG answers with inspectable sources; MCP tool audit records; and LLMOps traces/evaluation summaries. Water consumption and system uptime will be measured when physical test duration allows.

9. References

1. SUP'COM, *Cloud of Things: Scope Statement and Project Directives*, revised 8 October 2026 (course requirements).
2. Annabi, H. A., Laribi, B., & Bettaieb, T. (2025). "Cultivation of Genovese basil in aquaponic and hydroponic systems: A comparative study," *Journal of Agriculture and Applied Biology*, 6(2), 264–276. https://doi.org/10.11594/jaab.06.02.10.
3. WildFly Project (2026). "WildFly 42 is released!" https://www.wildfly.org/news/2026/10/22/WildFly-42-is-released/.
4. W3C (2023). *Web of Things Thing Description 1.1*. https://www.w3.org/TR/wot-thing-description11/.